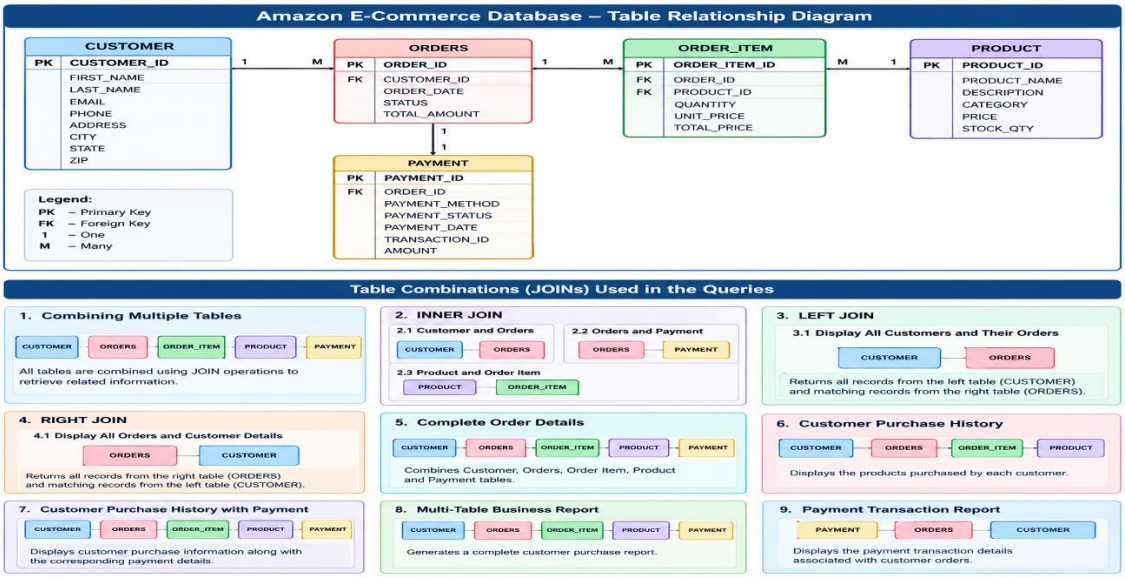
Amazon E-Commerce Database Management System

DATABASE RELATIONSHIP ANALYSIS USING JOINS

Chapter-9

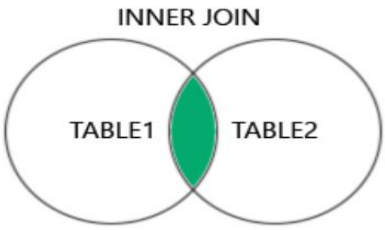
1. Combining Multiple Tables

The Amazon E-Commerce database contains different tables that store customer, product, order, and payment information. These tables can be combined using JOIN operations to retrieve related information.



2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.



INNER JOIN Syntax

SELECT *column_name(s)*
FROM *table1*
INNER JOIN *table2*
ON *table1.column_name* = *table2.column_name*;

2.1 Customer and Orders

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         O.ORDER_ID,
  5         O.ORDER_DATE,
  6         O.TOTAL_AMOUNT,
  7         O.ORDER_STATUS
  8  FROM CUSTOMER C
  9  INNER JOIN ORDERS O
 10  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  ORDER BY O.ORDER_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
         101 Arun
Kumar                                                 201 10-JAN-26
        1500 Delivered

         102 Priya
Sharma                                                202 11-JAN-26
        2500 Delivered

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------

         103 Rahul
Verma                                                 203 12-JAN-26
        1800 Cancelled

         104 Sneha
Reddy                                                 204 13-JAN-26
```

2.2 Orders and Payment

```
SQL> SELECT O.ORDER_ID,
  2         O.CUSTOMER_ID,
  3         O.ORDER_DATE,
  4         O.TOTAL_AMOUNT,
  5         P.PAYMENT_ID,
  6         P.PAYMENT_METHOD,
  7         P.PAYMENT_STATUS,
  8         P.AMOUNT
  9  FROM ORDERS O
 10  INNER JOIN PAYMENT P
 11  ON O.ORDER_ID = P.ORDER_ID
 12  ORDER BY O.ORDER_ID;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_ID
---------- ----------- ---------- ------------ ----------
PAYMENT_METHOD                 PAYMENT_STATUS           AMOUNT
------------------------------ -------------------- ----------
       201          101 10-JAN-26         1500         501
UPI                            Successful                1500

       202          102 11-JAN-26         2500         502
Credit Card                    Successful                2500

       203          103 12-JAN-26         1800         503
Debit Card                     Failed                    1800

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_ID
---------- ----------- ---------- ------------ ----------
PAYMENT_METHOD                 PAYMENT_STATUS           AMOUNT
------------------------------ -------------------- ----------
       204          104 13-JAN-26         3000         504
UPI                            Pending                   3000

       205          105 14-JAN-26         1200         505
Cash                           Successful                1200

       206          106 15-JAN-26         3500         506
```

2.3 Product and Order

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         P.PRICE,
  4         OI.ORDER_ID,
  5         OI.QUANTITY,
  6         OI.UNIT_PRICE
  7  FROM PRODUCT P
  8  INNER JOIN ORDER_ITEM OI
  9  ON P.PRODUCT_ID = OI.PRODUCT_ID
 10  ORDER BY OI.ORDER_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE   ORDER_ID   QUANTITY UNIT_PRICE
---------- ---------- ---------- ----------
       101
Laptop
     55000        201          1      55000

       102
Smartphone
     25000        202          1      25000

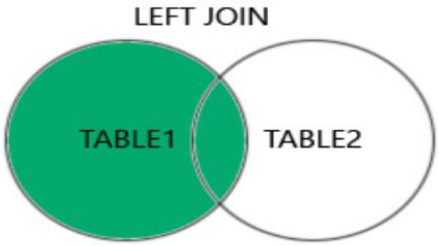
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE   ORDER_ID   QUANTITY UNIT_PRICE
---------- ---------- ---------- ----------

       103
Headphones
      1999        203          2       1999

       104
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



LEFT JOIN Syntax

SELECT *column_name(s)*
FROM *table1*
LEFT JOIN *table2*
ON *table1.column_name* = *table2.column_name*;

3.1 Display All Customers and Their Orders

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         O.ORDER_ID,
  5         O.ORDER_DATE,
  6         O.TOTAL_AMOUNT,
  7         O.ORDER_STATUS
  8  FROM CUSTOMER C
  9  LEFT JOIN ORDERS O
 10  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------
         101 Arun
Kumar                                                 201 10-JAN-26
        1500 Delivered

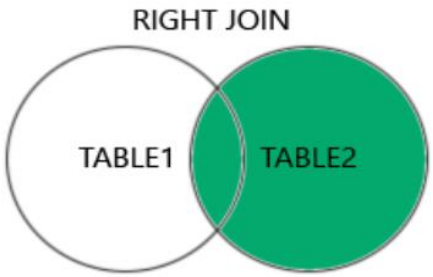
         101 Arun
Kumar                                                 208 17-JAN-26
        2800 Pending

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ --------------------

         102 Priya
Sharma                                                202 11-JAN-26
        2500 Delivered
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details

```
SQL> SELECT O.ORDER_ID,
  2         O.ORDER_DATE,
  3         O.TOTAL_AMOUNT,
  4         O.ORDER_STATUS,
  5         C.CUSTOMER_ID,
  6         C.FIRST_NAME,
  7         C.LAST_NAME
  8  FROM CUSTOMER C
  9  RIGHT JOIN ORDERS O
 10  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  ORDER BY O.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS          CUSTOMER_ID
---------- --------- ------------ -------------------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME
--------------------------------------------------
       201 10-JAN-26         1500 Delivered                    101
Arun
Kumar

       202 11-JAN-26         2500 Delivered                    102
Priya
Sharma

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS          CUSTOMER_ID
---------- --------- ------------ -------------------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME
--------------------------------------------------

       203 12-JAN-26         1800 Cancelled                    103
Rahul
Verma
```

5. Complete Order Details

This query combines Customer, Orders, Order Item, Product, and Payment tables.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         O.ORDER_ID,
  5         O.ORDER_DATE,
  6         O.ORDER_STATUS,
  7         P.PRODUCT_ID,
  8         P.PRODUCT_NAME,
  9         OI.QUANTITY,
 10         OI.UNIT_PRICE,
 11         PY.PAYMENT_METHOD,
 12         PY.PAYMENT_STATUS,
 13         PY.AMOUNT
 14  FROM CUSTOMER C
 15  INNER JOIN ORDERS O
 16  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 17  INNER JOIN ORDER_ITEM OI
 18  ON O.ORDER_ID = OI.ORDER_ID
 19  INNER JOIN PRODUCT P
 20  ON OI.PRODUCT_ID = P.PRODUCT_ID
 21  INNER JOIN PAYMENT PY
 22  ON O.ORDER_ID = PY.ORDER_ID
 23  ORDER BY O.ORDER_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS          PRODUCT_ID
--------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- -------------------------------- --------------------
    AMOUNT
```

```
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- -------------------------------- --------------------
    AMOUNT
----------
Delivered                   101

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS          PRODUCT_ID
--------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- -------------------------------- --------------------
    AMOUNT
----------
Laptop

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
ORDER_STATUS          PRODUCT_ID
--------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- -------------------------------- --------------------
    AMOUNT
----------
         1      55000 UPI                              Successful

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
```

6. Customer Purchase History

This query displays the products purchased by each customer.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         O.ORDER_ID,
  5         P.PRODUCT_ID,
  6         P.PRODUCT_NAME,
  7         OI.QUANTITY,
  8         OI.UNIT_PRICE
  9  FROM CUSTOMER C
 10  INNER JOIN ORDERS O
 11  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 12  INNER JOIN ORDER_ITEM OI
 13  ON O.ORDER_ID = OI.ORDER_ID
 14  INNER JOIN PRODUCT P
 15  ON OI.PRODUCT_ID = P.PRODUCT_ID
 16  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID PRODUCT_ID
-------------------------------------------------- ---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE
---------- ----------
       101 Arun
Kumar                                                   201        101
Laptop
         1      55000


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID PRODUCT_ID
-------------------------------------------------- ---------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
```

7. Customer Purchase History with Payment

This query displays customer purchase information along with the corresponding payment details.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         O.ORDER_ID,
  5         P.PRODUCT_NAME,
  6         OI.QUANTITY,
  7         OI.UNIT_PRICE,
  8         PY.PAYMENT_METHOD,
  9         PY.PAYMENT_STATUS,
 10         PY.AMOUNT
 11  FROM CUSTOMER C
 12  INNER JOIN ORDERS O
 13  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 14  INNER JOIN ORDER_ITEM OI
 15  ON O.ORDER_ID = OI.ORDER_ID
 16  INNER JOIN PRODUCT P
 17  ON OI.PRODUCT_ID = P.PRODUCT_ID
 18  INNER JOIN PAYMENT PY
 19  ON O.ORDER_ID = PY.ORDER_ID
 20  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- ------------------------------ --------------------
    AMOUNT
----------
         101 Arun
Kumar                                                   201
Laptop

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID
```

8. Multi-Table Business Report

The following query generates a complete customer purchase report.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.FIRST_NAME,
  3         C.LAST_NAME,
  4         O.ORDER_ID,
  5         P.PRODUCT_NAME,
  6         OI.QUANTITY,
  7         OI.UNIT_PRICE,
  8         O.TOTAL_AMOUNT,
  9         PY.PAYMENT_METHOD,
 10         PY.PAYMENT_STATUS
 11  FROM CUSTOMER C
 12  INNER JOIN ORDERS O
 13  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 14  INNER JOIN ORDER_ITEM OI
 15  ON O.ORDER_ID = OI.ORDER_ID
 16  INNER JOIN PRODUCT P
 17  ON OI.PRODUCT_ID = P.PRODUCT_ID
 18  INNER JOIN PAYMENT PY
 19  ON O.ORDER_ID = PY.ORDER_ID
 20  ORDER BY O.ORDER_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE TOTAL_AMOUNT PAYMENT_METHOD
---------- ---------- ------------ ------------------------------
PAYMENT_STATUS
--------------------
         101 Arun
Kumar                                                   201
Laptop

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
```

```
  QUANTITY UNIT_PRICE TOTAL_AMOUNT PAYMENT_METHOD
---------- ---------- ------------ ------------------------------
PAYMENT_STATUS
--------------------
         1      55000         1500 UPI
Successful


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE TOTAL_AMOUNT PAYMENT_METHOD
---------- ---------- ------------ ------------------------------
PAYMENT_STATUS
--------------------
       102 Priya
Sharma                                                 202
Smartphone

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE TOTAL_AMOUNT PAYMENT_METHOD
---------- ---------- ------------ ------------------------------
PAYMENT_STATUS
--------------------
         1      25000         2500 Credit Card
Successful


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
```

9. Payment Transaction Report

This report displays the payment transaction details associated with customer orders.

```
SQL> SELECT PY.PAYMENT_ID,
  2         PY.ORDER_ID,
  3         C.CUSTOMER_ID,
  4         C.FIRST_NAME,
  5         C.LAST_NAME,
  6         PY.PAYMENT_METHOD,
  7         PY.PAYMENT_STATUS,
  8         PY.PAYMENT_DATE,
  9         PY.TRANSACTION_ID,
 10         PY.AMOUNT
 11  FROM PAYMENT PY
 12  INNER JOIN ORDERS O
 13  ON PY.ORDER_ID = O.ORDER_ID
 14  INNER JOIN CUSTOMER C
 15  ON O.CUSTOMER_ID = C.CUSTOMER_ID
 16  ORDER BY PY.PAYMENT_ID;

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- ------------
FIRST_NAME
--------------------------------------------------
LAST_NAME
--------------------------------------------------
PAYMENT_METHOD                 PAYMENT_STATUS       PAYMENT_D
------------------------------ -------------------- ---------
TRANSACTION_ID                      AMOUNT
------------------------------ ----------
       501        201         101
Arun
Kumar

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- ------------
FIRST_NAME
--------------------------------------------------
LAST_NAME
--------------------------------------------------
PAYMENT_METHOD                 PAYMENT_STATUS       PAYMENT_D
------------------------------ -------------------- ---------
TRANSACTION_ID                      AMOUNT
```